

Veritype

One-tap fact-checking inside Slack,
with chart-rich verdict cards rendered
through the Datatype variable font.



Version 0.1.0
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Track New Slack Agent
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What's inside.

A practical guide for installing, running, demoing, and extending the Veritytype Slack agent.

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01 THE PROBLEM

Misinformation moves faster than the team can Google.

Slack is the substrate that modern teams use to trade information at speed. That speed is also why half-true claims spread before anyone has time to challenge them.

The person who eventually pushes back has to do their homework alone, in a side tab, then disrupt the thread to share it. By that point the original claim has been internalised by three other people and re-shared in a different channel.

What today looks like

WITHOUT VERITYTYPE

- Claim drops in #general at 09:47.
- Three emoji-reactions, no challenge.
- 10:30, someone tries to push back. Has to Google for ten minutes.
- Source debate derails the channel for the rest of the morning.
- The team's track record on disputed claims is invisible.

WITH VERITYTYPE

- Bot offers a one-tap verification, privately to the speaker.
- Verdict arrives in 5 to 8 seconds with weighted citations.
- Card stays in thread. Sources are one click away.
- Channel stays calm, the work resumes.
- Workspace dashboard tracks claims, verdicts, latency, coverage.

Why a fact-checking agent, not a fact-checking link

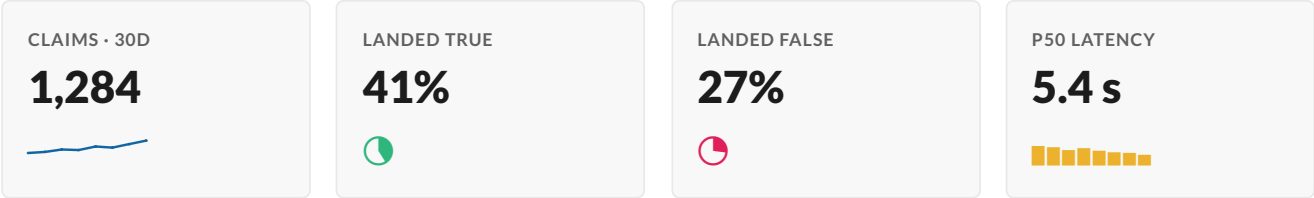
People do not stop to verify claims when the cost is opening another tab. People do stop to verify when the cost is a single button. Slack apps that are not in-channel get used twice and forgotten. Veritytype lives in the conversation, with the same APP pill any other bot has, and never makes you context-switch.

Why a font for the chart? A PNG chart is cute. A chart that is also valid text inside a Datatype-styled card is searchable, copy-pasteable, scales with text size, and gives the verdict an inarguably unique visual identity. Nobody else in this hackathon is doing it.

02 WHAT IT DOES

Listen, verify, render.

Three small steps. One feels like the bot is paying attention. One feels like the bot is doing real work. One feels like the bot has visual taste.



The pipeline

STEP	WHAT HAPPENS	MODEL
Listen	Every channel message hits a triage classifier that decides whether it asserts a verifiable, public-interest claim. If yes, an ephemeral "want me to verify this?" prompt appears.	<code>c laude-haiku-4-5</code>
Verify	On accept, a research agent fans out web searches, weighs sources by quality, and synthesises a verdict (TRUE / MOSTLY TRUE / MIXED / MISLEADING / FALSE / UNVERIFIED) with confidence, citations, and context.	<code>c laude-opus-4-7</code> + <code>web_search</code>
Render	The verdict is painted into a Datatype-font HTML card and rasterised by Puppeteer. Slack thread gets a PNG plus a Block Kit mrkdwn fallback for accessibility, search, and mobile.	Puppeteer + Datatype

Three Datatype expressions, three insights

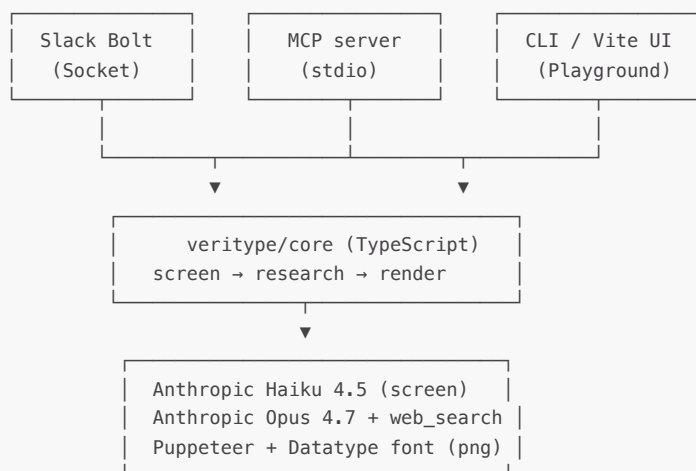
The font turns text like `{b:20,45,70}` into a real bar chart through ligature substitution. Every verdict card uses three of these:

- `{b:...}` source-weight bars: primary documents sit taller than blog posts.
- `{p:...}` three pies for supports / refutes / context as evidence balance.
- `{l:...}` a sparkline of how often the claim has been showing up over time.

03 ARCHITECTURE

One brain, three surfaces.

Everything lives in `src/`. The Slack bot, the MCP server, and the demo dashboard are three thin shells around a single core module.



Why split the brain from the bot

If the fact-checking logic only lived inside the Slack handler, a teammate using Claude Desktop could not call it. By isolating it in `src/core` and re-exposing it through MCP, the same verdict that lands in a Slack thread can be requested from any MCP-aware client.

Hackathon requirement check. Slack Agent Builder Challenge requires at least one of: Slack AI capabilities, MCP server integration, Real-Time Search API. Veritytype hits two: a real MCP server and Anthropic's `web_search_20250305` tool used by every verification.

Failure modes built in

- If Opus is unavailable or rate-limited, the researcher falls back to `claude-sonnet-4-6` with the same JSON contract.
- If the screener returns malformed JSON, the message is silently skipped (no offer, no error pollution).
- If Puppeteer fails to render, the bot still posts the mrkdwn Block Kit fallback so users always see the verdict.

04 COMPONENTS

Every file that does something interesting.

COMPONENT	ROLE	PATH
Slack Bolt app	Socket-mode app, <code>/verify</code> slash command, message events, button handlers.	<code>src/slack/app.ts</code>
Block Kit blocks	Verdict block layout, mrkdwn fallback, action buttons.	<code>src/slack/blocks.ts</code>
Claim screener	Haiku 4.5 layer that decides whether a message contains a verifiable claim.	<code>src/core/screener.ts</code>
Researcher	Opus 4.7 + <code>web_search</code> agent. Returns verdict, confidence, weighted sources.	<code>src/core/researcher.ts</code>
Card template	Datatype-styled HTML used by both the PNG renderer and the dashboard.	<code>src/core/cardTemplate.ts</code>
Puppeteer renderer	Rasterises HTML cards to PNG with hi-DPI scale factor.	<code>src/core/renderer.ts</code>
MCP server	Exposes <code>screen_message</code> , <code>verify_claim</code> , <code>render_verdict_card</code> over stdio MCP.	<code>src/mcp/server.ts</code>
Demo dashboard	Slack-skinned Vite/React UI for screenshots, library browsing, playground.	<code>src/web/</code>
Datatype font	The variable font that makes <code>{b:...}</code> , <code>{l:...}</code> , <code>{p:...}</code> render as charts.	<code>assets/fonts/</code>
Sample fixtures	Three pre-baked reports used by the playground and CLI renderer.	<code>src/core/sample.ts</code>

Built with

TypeScript

Node.js 22

Slack Bolt

Slack Block Kit

Anthropic SDK

Claude Haiku 4.5

Claude Opus 4.7

Anthropic web_search

Model Context Protocol

Puppeteer

Datatype font

Vite

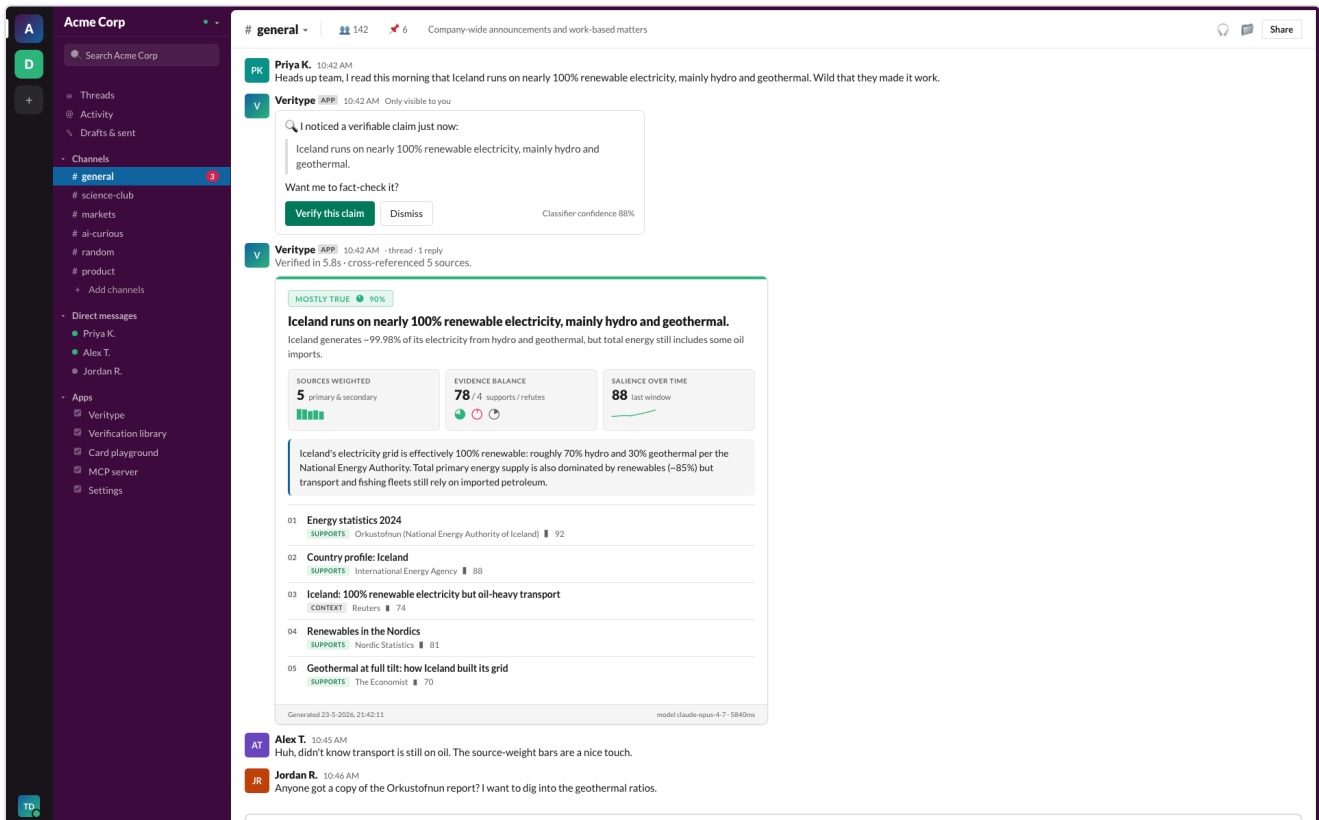
React 18

python-pptx

05 IN-CHANNEL UX

The proactive offer.

Veritype's most distinctive UX move: it asks first. Every message in a watched channel goes through the Haiku screener, and when a verifiable claim is detected, the bot drops an ephemeral prompt visible only to the speaker. No public noise, no shaming.



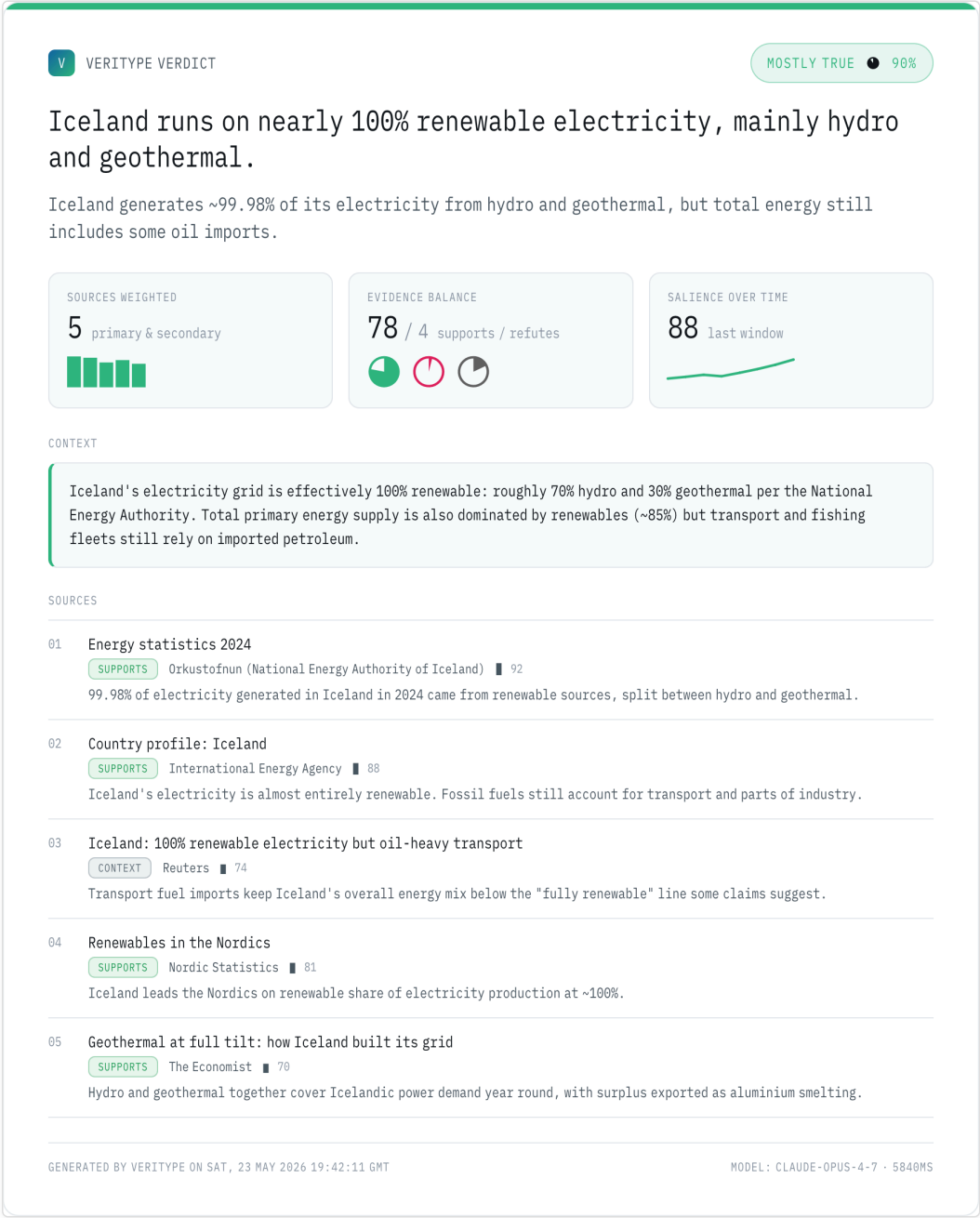
Three properties that make it feel native

1. **Ephemeral by default.** The offer is only seen by the message author. If they dismiss, nobody else ever knew.
2. **Threaded by default.** Verdicts post in thread so the main channel stays scannable.
3. **Block Kit + image dual surface.** The card is sent both as a Datatype PNG and as native Block Kit mrkdwn, so it works on mobile, screen readers, and Slack search.

06 THE VERDICT CARD

Anatomy of a Datatype card.

Every verdict card is the same template, rendered with the verdict-specific accent color. The three Datatype expressions across the metrics row tell you the whole story without reading a single sentence.

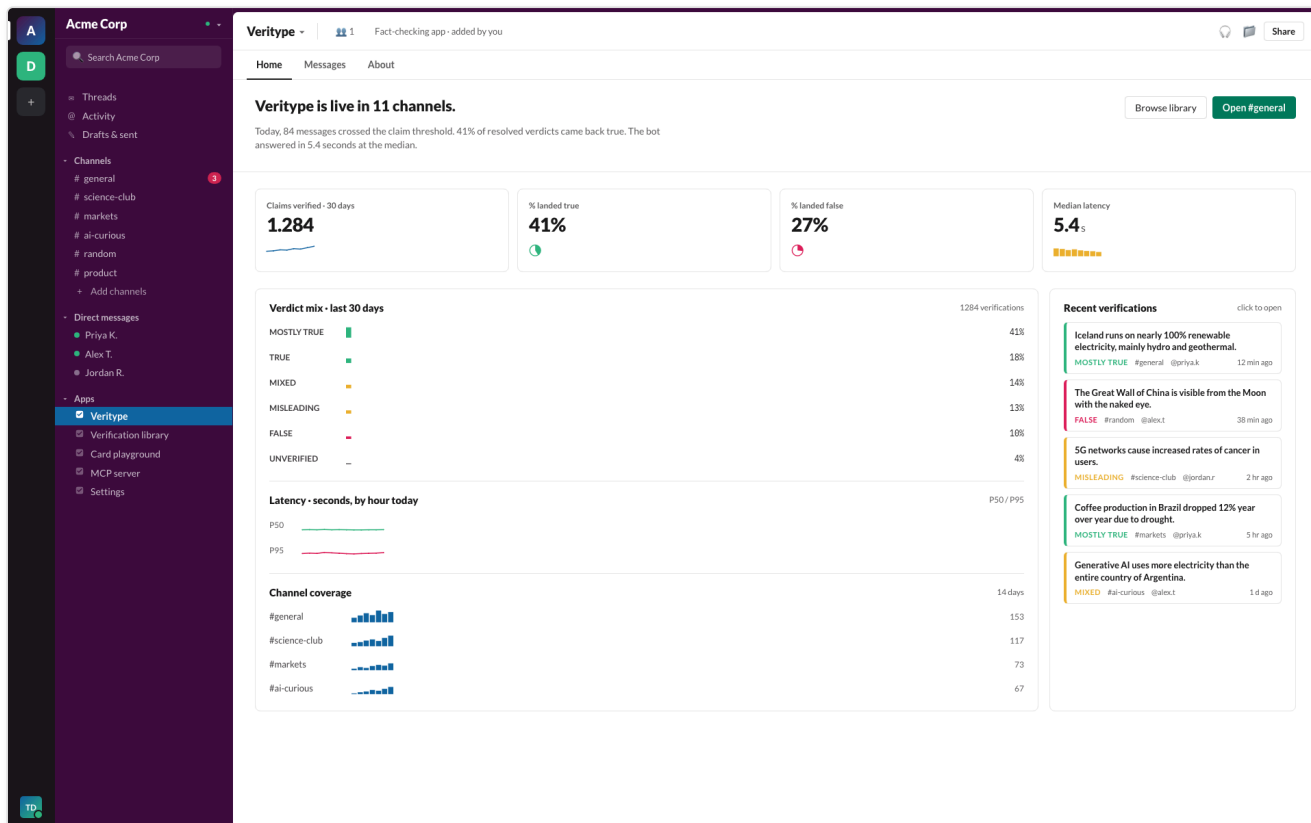


Reading the card

ELEMENT	TELLS YOU
Verdict pill	TRUE / MOSTLY TRUE / MIXED / MISLEADING / FALSE / UNVERIFIED with a confidence pie and percentage.
Claim line Veritype · 0.1.0	The standardised one-sentence rephrasing the agent verified, not the raw message.
One-liner	The bottom line in 140 characters, no hedging.

Admins get a Slack-style dashboard.

Open the Veritype app in Slack and you land on its Home tab. KPIs render through Datatype, verdict mix, latency, and channel coverage charts are all live. Recent verifications are one click away.



What admins look at first

- **Claim volume:** tells you whether the bot is being used at all, and whether usage is growing.
- **True / False split:** an early signal of culture. A workspace where most claims land false is making different bets than one where most land true.
- **Latency:** P50 should sit under 7 seconds. If it creeps up, the web_search budget is probably too high.
- **Channel coverage:** which channels are over-listened (consider scoping the regex) and which are silent (consider inviting the bot).

08 VERIFICATION LIBRARY

Every claim, every citation, searchable.

The Library tab is the long-tail memory of the bot. Every verification, across every channel, with the full source list intact.

Veritype - 1 Verification library - all workspaces

Home Library Messages About

Verification library

Every claim Veritype has fact-checked across your workspace, with full citations. Click any verdict to open the card.

Search claims... Export CSV

MOSTLY TRUE 90%

Iceland runs on nearly 100% renewable electricity, mainly hydro and geothermal.

#general @priya.k 12 min ago 5 sources

FALSE

The Great Wall of China is visible from the Moon with the naked eye.

#random @alex.t 38 min ago 4 sources

MISLEADING

5G networks cause increased rates of cancer in users.

#science-club @jordan.r 2 hr ago 4 sources

MOSTLY TRUE

Coffee production in Brazil dropped 12% year over year due to drought.

#markets @priya.k 5 hr ago 4 sources

MIXED

Generative AI uses more electricity than the entire country of Argentina.

#ai-curious @alex.t 1 d ago 4 sources

FALSE

The 2024 Nobel Prize in Physics went to a researcher in cold fusion.

#science-club @jordan.r 1 d ago 4 sources

MOSTLY TRUE 90%

Iceland runs on nearly 100% renewable electricity, mainly hydro and geothermal.

Iceland generates ~99.98% of its electricity from hydro and geothermal, but total energy still includes some oil imports.

SOURCES WEIGHTED: 5 primary & secondary

EVIDENCE BALANCE: 78 / 4 supports / refutes

SALIENCE OVER TIME: 88 last window

Iceland's electricity grid is effectively 100% renewable: roughly 70% hydro and 30% geothermal per the National Energy Authority. Total primary energy supply is also dominated by renewables (~85%) but transport and fishing fleets still rely on imported petroleum.

- Energy statistics 2024
SUPPORTS: Orkustofnun (National Energy Authority of Iceland) 92
- Country profile: Iceland
SUPPORTS: International Energy Agency 88
- Iceland: 100% renewable electricity but oil-heavy transport
CONTEXT: Reuters 74
- Renewables in the Nordics
SUPPORTS: Nordic Statistics 81
- Geothermal at full tilt: how Iceland built its grid
SUPPORTS: The Economist 70

Generated 23-5-2024, 21:42:11 model: claude-opus-4-7-5840ms

Use it for

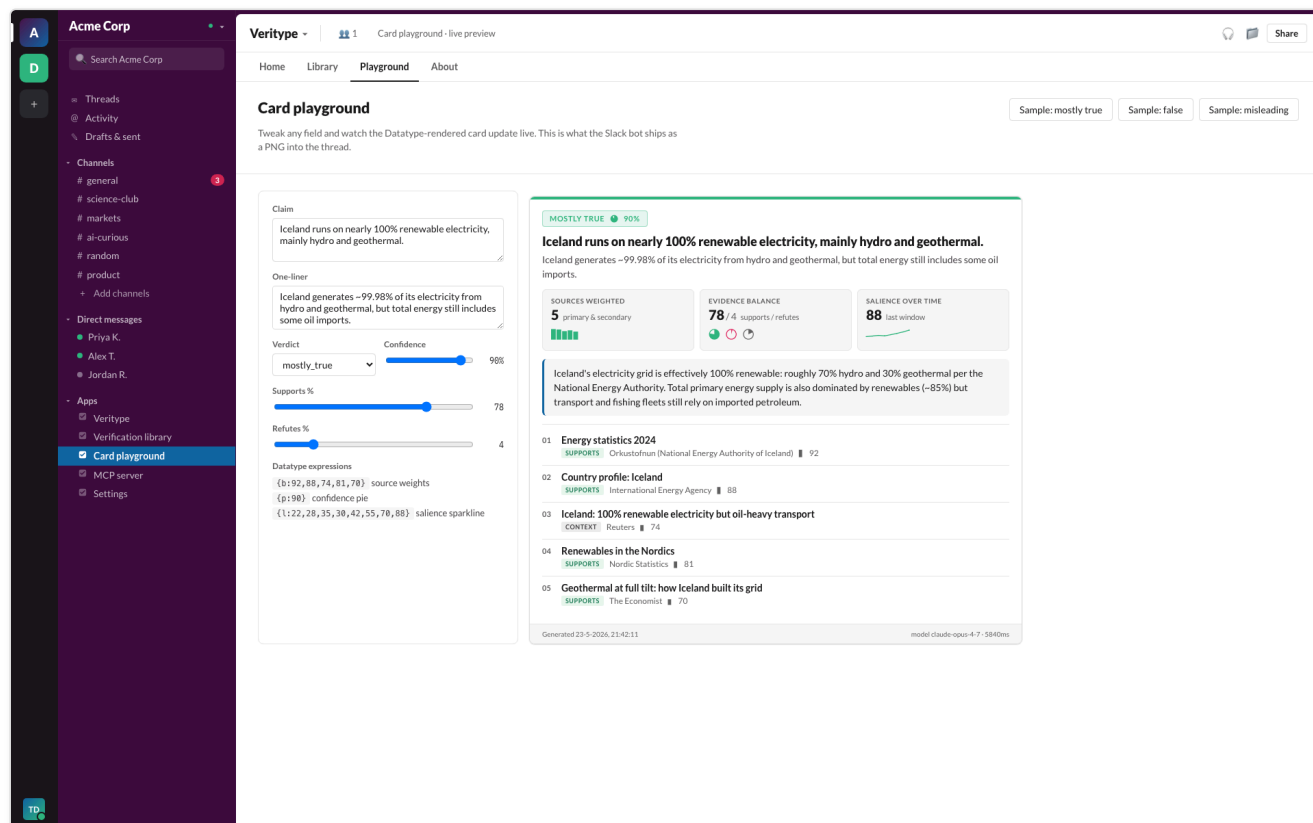
- **Recurring debates.** "We already verified this in March. Here's the card."
- **Onboarding.** New hires can browse the team's litigated facts to learn the house view.
- **Audit.** If a claim's status changes (a study is retracted, a number is revised), re-verify and the library tracks the delta.

09 PLAYGROUND & MCP

Edit live. Plug in anywhere.

Card playground

Tweak any field of a sample report and watch the Datatype card re-render. Great for visually testing new verdict types or alternate copy before shipping.



MCP server

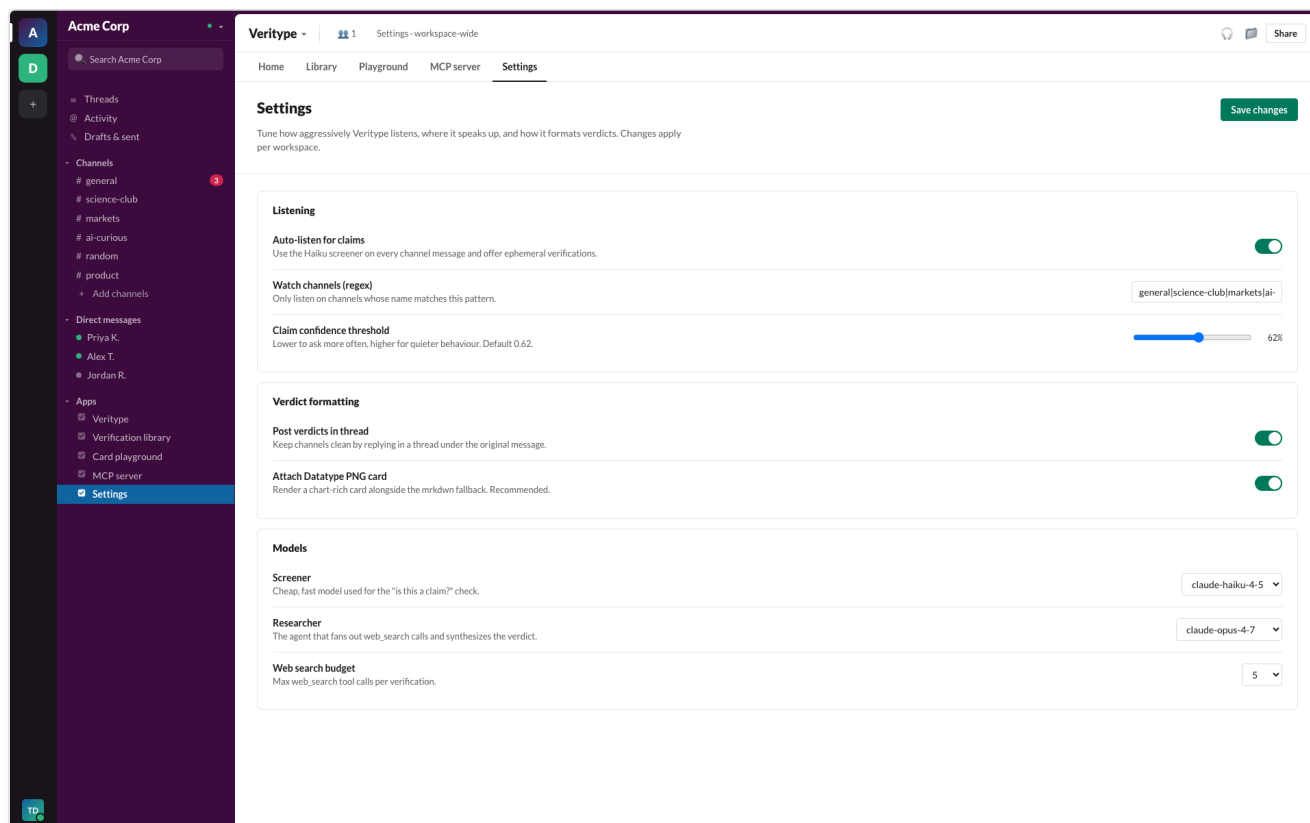
Drop this into your Claude Desktop config and the three Veritype tools become callable from any MCP-aware client:

```
// ~/.claude/claude_desktop_config.json
{
  "mcpServers": {
    "veritype": {
      "command": "node",
      "args": ["/path/to/veritype/dist/mcp/server.js"],
      "env": { "ANTHROPIC_API_KEY": "sk-ant-..." }
    }
  }
}
```

10 SETTINGS

Tune the listener. Trust the verdict.

Veritype is meant to be quiet by default. The Settings tab is where you make it quieter, louder, or more discerning.



Knobs that matter

SETTING	WHEN TO ADJUST
Auto-listen	Turn off if the team prefers explicit <code>/verify</code> only. On is the default.
Watch channels regex	Use to whitelist channels. Common pattern: <code>general product markets</code> .
Confidence threshold	Default 0.62. Raise to 0.75+ for very quiet channels; lower to 0.45 for high-signal places like <code>#science-club</code> .
Post in thread	Default on. Off is for channels where the team wants verdicts in-line.
Attach PNG card	Default on. Off only if you have a workspace-wide rule against image attachments.
Researcher model	Opus 4.7 is the default. Sonnet 4.6 is the fallback and is ~30% cheaper.
Web-search budget	5 is a good ceiling. 3 for cost-sensitive workspaces, 8+ for very contested claims.

11 SETUP

From cold clone to live workspace.

Prerequisites

- Node.js 20+
- An Anthropic API key with access to Opus 4.7 and the `web_search_20250305` tool.
- A Slack workspace where you can install custom apps.

Install and configure

```
# 1. clone and install
git clone https://github.com/timdries/veritype.git
cd veritype
npm install

# 2. configure env
cp .env.example .env
# fill in ANTHROPIC_API_KEY, SLACK_BOT_TOKEN, SLACK_APP_TOKEN, SLACK_SIGNING_SECRET
```

Create the Slack app

1. Go to api.slack.com/apps and click **Create New App** → **From a manifest**.
2. Paste `docs/slack-app-manifest.yaml`.
3. Enable Socket Mode and generate an app-level token with `connections:write`.
4. Install to your workspace.

Run

```
# start the Slack agent (Socket Mode)
npm run dev:slack

# start the demo dashboard (optional)
npm run dev:web
```

Type `/verify Iceland` runs on 100% renewable electricity in any channel the bot is a member of. The verdict card lands in thread within ~6 seconds.

Common gotchas. Puppeteer downloads Chromium on first `npm install`; if behind a corporate proxy set `PUPPETEER_DOWNLOAD_BASE_URL`. If the bot is silent, double-check the bot is invited to the channel *and* that the channel name matches the `VERITYPE_WATCH_CHANNELS` regex.

12 RESOURCES

Where to look next.

Links

Repo	github.com/timdries/veritytype
Devpost	slackhack.devpost.com
Demo video	youtu.be/REPLACE_WITH_DEMO_ID
Datatype font	franktisellano.github.io/datatype
Slack Bolt	slack.dev/bolt-js
Anthropic web_search	docs.anthropic.com/en/docs/build-with-claude/tool-use/web-search-tool
Model Context Protocol	modelcontextprotocol.io

Repo layout

```
src/  
  core/          shared brain (screener, researcher, renderer, types, samples)  
  slack/         Bolt app + Block Kit blocks  
  mcp/           stdio MCP server wrapping the same core  
  web/           Vite + React dashboard (Slack-skinned, demo + screenshots)  
assets/  
  fonts/         Datatype woff2 (Regular / Medium / SemiBold / Bold)  
docs/  
  screenshots/   in-app screenshots, sample verdict cards  
  manual/        operator manual sources + rendered PDF  
deck/            pitch deck (.pptx + .pdf)
```

License

The Veritytype code is MIT. The Datatype font is Apache 2.0; see [franktisellano/datatype](https://franktisellano.github.io/datatype) for the source and full license text.

Verify in channel. Trust on sight.

Veritype is the Slack-native fact-checker for teams that move fast on shared information. Coded agent, MCP-ready, chart-rich, ephemeral when it should be, in-thread when it counts.

The repo, the demo, the deck, and this manual are all linked from the README. If anything is unclear, open an issue or DM Tim.

Slack-native fact-checking, with charts that ride
inside type.